MAL

RADICAVA®

edaravone

Mitsubishi Tanabe Pharma

1. NAME OF THE MEDICINE PRODUCT

RADICAVA® IV concentrate solution for infusion 30mg/20ml

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

(INN: Edaravone)

Each ampoule contains 30mg of edaravone in 20mL. After dilution, 1mL of solution contains approximately 0.43mg of edaravone.

RADICAVA contains sodium bisulfite, a sulfite that may cause allergic type reactions. For the full list of excipients, see *List of excipients (section 6.1).*

3. PHARMACEUTICAL FORM

Injection
A clear and colourless aqueous solution for injection.

4. CLINICAL PARTICULARS

4.1 Therapeutic indication

RADICAVA (edaravone) is indicated to slow the loss of function in patients with amyotrophic lateral sclerosis (ALS).

4.2 Posology and mode of administration

Posology
The usual adult dosage is two ampoules (60 mg of edaravone) diluted with 100mL of physiological saline, etc., which is administered intravenously over 60 minutes once a day.

Special population
Paediatric Use
Safety and effectiveness of RADICAVA in paediatric patients have not been established.

Geriatric Use
Of the 184 patients with ALS who received RADICAVA in 3 placebo-controlled clinical trials, a total of 53 patients were 65 years of age and older, including 2 patients 75 years of age and older. No overall differences in safety or effectiveness were observed between these patients and younger patients, but greater sensitivity of some older individuals cannot be ruled out.

Renal Impairment
Mild and moderate renal impairment have no clinically significant effects on the pharmacokinetics of edaravone. No dose adjustment is needed in these patients. No studies have been performed to characterize the pharmacokinetics of edaravone or establish its safety in patients with severe renal impairment.

Hepatic Impairment
Mild, moderate and severe hepatic impairment have no clinically significant effects on the pharmacokinetics of edaravone. No dose adjustment is needed in these patients.

Mode of Administration
Administer two ampoules (60mg of edaravone) diluted in 100 mL of physiological saline, over a total of 60 minutes (infusion rate approximately 1 mg per minute [2.33 mL per minute]).

Usually, the duration of administration and cessation of this product are combined in one cycle of treatment for 28 days and the cycle should be repeated. This product is consecutively infused for 14 days in the duration of administration followed by cessation for 14 days in the 1st cycle, and from the 2nd cycle, this product is infused for 10 of 14 days in the duration of administration followed by cessation for 14 days.

Promptly discontinue the infusion upon the first observation of any signs or symptoms consistent with a hypersensitivity reaction [see *Special warnings and precautions for use (section 4.4)*].

Other medications should not be mixed with RADICAVA.

4.3 Contraindications

RADICAVA is contraindicated in patients with a history of hypersensitivity to edaravone or to any of the ingredients of RADICAVA [see *List of excipients (section 6.1)*].

4.4 Special warnings and precautions for use

General

Hypersensitivity Reactions

Hypersensitivity reactions (redness, wheals, and erythema multiforme) and cases of anaphylaxis (urticaria, decreased blood pressure, and dyspnea) have been reported in spontaneous post marketing reports with RADICAVA.

Patients should be monitored carefully for hypersensitivity reactions. If hypersensitivity reactions occur, discontinue RADICAVA, treat per standard of care, and monitor until the condition resolves [*see Contraindications (section 4.3)*].

Sulfite Allergic Reactions

RADICAVA contains sodium bisulfite, a sulfite that may cause allergic type reactions, including anaphylactic symptoms and life-threatening or less severe asthmatic episodes in susceptible people. The overall prevalence of sulfite sensitivity in the general population is unknown. Sulfite sensitivity occurs more frequently in asthmatic people.

Precautions when opening the ampoule

RADICAVA is supplied in a "one-point-cut ampoule". Break the ampoule while pulling its neck downward with the round mark frontal. To avoid contamination with foreign substances upon cutting ampoule, the cut point of the ampoule should be wiped with an alcohol swab before opening.

4.5 Interaction with other medicinal products and other forms of interaction

The pharmacokinetics of edaravone is not expected to be significantly affected by inhibitors of CYP enzymes, UGTs, or major transporters.

In vitro studies demonstrated that, at the recommended clinical dose, edaravone and its metabolites are not expected to significantly inhibit cytochrome P450 enzymes (CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6, CYP3A4), UGT1A1, UGT2B7, or transporters (P-gp, BCRP, OATP1B1, OATP1B3, OAT1, OAT3, and OCT2) in humans. Edaravone and its metabolites are not expected to induce CYP1A2, CYP2B6, or CYP3A4 at the recommended clinical dose of RADICAVA.

4.6 Fertility, pregnancy and lactation

Pregnancy
There are no adequate data on the developmental risk associated with the use of RADICAVA in pregnant women. In animal studies, administration of edaravone to pregnant rats and rabbits resulted in adverse developmental effects (increased mortality, decreased growth, delayed sexual development, and altered behaviour) at clinically relevant doses. Most of these effects occurred at doses that were also associated with maternal toxicity [see *Preclinical safety data (section 5.4)*].

Lactation
There are no data on the presence of edaravone in human milk, the effects on the breastfed infant, or the effects of the drug on milk production. However, edaravone and its metabolites are excreted in the milk of lactating rats. Because many drugs are excreted in human milk, caution should be exercised. The developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for RADICAVA and any potential adverse effects on the breastfed infant from RADICAVA or from the underlying maternal condition.

4.7 Effects on ability to drive and use machines

No studies on the effect of RADICAVA on the ability to drive and use machines have been performed.

4.8 Undesirable effects

4.8.1 Overview of undesirable effects

The following serious adverse reactions are described elsewhere in the labeling:
- Hypersensitivity Reactions [see *Special warnings and precautions for use (section 4.4)*]
- Sulfite Allergic Reactions [see *Special warnings and precautions for use (section 4.4)*]

4.8.2 Undesirable effects from clinical trial

Because clinical trials are conducted under very specific conditions, the adverse reaction rates observed in the clinical trials may not reflect the rates observed in practice and should not be compared to the rates in the clinical trials of another drug. Adverse reaction information from clinical trials is useful for identifying drug-related adverse events and for approximating rates.

The safety profile of RADICAVA compared to placebo was assessed in 3 clinical trials:
- Two double-blind, randomized, placebo-controlled studies in patients with grade 1-2 ALS (Japanese severity classification) with a total 343 subjects
- One double-blind, randomized, placebo-controlled study in patients with grade 3 ALS with 25 subjects

In the double-blind randomized, placebo-controlled trials, 184 ALS patients were administered RADICAVA 60 mg or placebo in treatment cycles for 6 months. The population consisted of Japanese patients who had a median age of 60 years (range 29-75) and were 59% male. Most (93%) of these patients were living independently at the time of screening.

Table 1 lists the adverse reactions that occurred in ≥ 2% of patients in the RADICAVA-treated group and that occurred at least 2% more frequently than in the placebo-treated group in randomized placebo-controlled ALS trials. The most common adverse reactions that occurred in ≥10% of RADICAVA-treated patients were contusion, gait disturbance, and headache.

Table 1: Adverse Reactions from Pooled Placebo-Controlled Trials[a] that Occurred in 2% of RADICAVA-Treated Patients and 2% More Frequently than in Placebo Patients

Adverse Reaction	RADICAVA (N=184) %	Placebo (N=184) %
Contusion	15	9

Gait disturbance	13	9
Headache	10	6
Dermatitis	8	5
Eczema	7	4
Respiratory failure, respiratory disorder, hypoxia	6	4
Glycosuria	4	2
Tinea infection	4	2

[a] Pooled placebo-controlled studies include two additional studies with 231 additional patients, all using the same treatment regimen [see *Clinical Trials (section 5.2.1)*].

4.8.3 Post-market undesirable effects

The following adverse reactions have been identified during post-approval use of RADICAVA outside of Malaysia. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

Skin and subcutaneous tissue disorders: Hypersensitivity reactions and anaphylaxis.

4.8 Overdose

For management of a suspected drug overdose, contact your national poison control centre.

5 PHARMACOLOGICAL PROPERTIES

5.1 Mechanism of Action

The mechanism by which edaravone exerts its therapeutic effect in patients with ALS is unknown.

5.2 Pharmacodynamic properties

Cardiac Electrophysiology: In a Japanese study in healthy male subjects, at a dose 5 times the recommended dose, RADICAVA did not prolong the QT interval to any clinically relevant extent.

5.2.1 Clinical Trials

5.2.1.1 Trial Design and Study Demographics

Table 2: Summary of patient demographics for clinical trials in Amyotrophic Lateral Sclerosis

Study #	Trial design	Dosage, route of administration and duration	Study subjects (n)	Mean age (Range)	Sex
MCI186-19	Randomized, placebo-controlled, double-blind study	Intravenous infusion of 60 mg over 60 minutes, 6 months (6 cycles) treatment	Edaravone (69) Placebo (68)	60.5 (30-75)	Male (55%) Female (45%)

5.2.1.2 Study Results

Study MCI186-19

The efficacy of RADICAVA for the treatment of ALS was established in a 6-month, randomized, placebo-controlled, double-blind study conducted in Japanese patients with ALS who were living independently and met the following criteria at screening:

1. Functionality retained in most activities of daily living (defined as scores of 2 points or better on each individual item of the ALS Functional Rating Scale – Revised [ALSFRS- R; described below])
2. Normal respiratory function (defined as percent-predicted forced vital capacity values of ≥ 80%)
3. Definite or Probable ALS based on El Escorial revised criteria
4. Disease duration of 2 years or less

The study enrolled 69 patients in the RADICAVA arm and 68 in the placebo arm. Baseline characteristics were similar between these groups, with over 90% of patients in each group being treated with riluzole.

RADICAVA was administered as an intravenous infusion of 60 mg given over a 60 minutes period according to the following schedule:

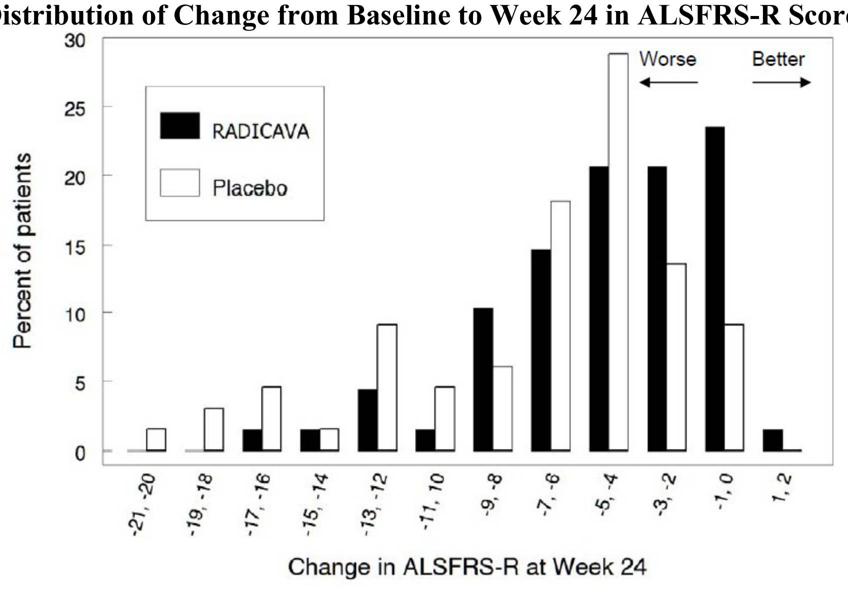
- An initial treatment cycle with daily dosing for 14 days, followed by a 14-day drug-free period (Cycle 1)
- Subsequent treatment cycles with daily dosing for 10 days out of 14-day periods, followed by 14-day drug-free periods (Cycles 2-6).

The primary efficacy endpoint was a comparison of the change between treatment arms in the ALSFRS-R total scores from baseline to Week 24. The ALSFRS-R scale consists of 12 questions that evaluate the fine motor, gross motor, bulbar, and respiratory function of patients with ALS (speech, salivation, swallowing, handwriting, cutting food, dressing/hygiene, turning in bed, walking, climbing stairs, dyspnea, orthopnea, and respiratory insufficiency). Each item is scored from 0-4, with higher scores representing greater functional ability. The decline in ALSFRS-R scores from baseline was significantly less in the RADICAVA-treated patients as compared to placebo (see Table 3). The distribution of change in ALSFRS-R scores from baseline to Week 24 by percent of patients is shown in Figure 1.

Table 3: Analysis of Change from Baseline to Week 24 in ALSFRS-R Scores

Treatment	Change from Baseline LS Mean ± SE (95% CI)	Treatment Difference (RADICAVA – placebo [95% CI])	*p*-value
RADICAVA 60mg	−5.01±0.64	2.49 (0.99, 3.98)	0.0013
Placebo	−7.50±0.66		

Figure 1: Distribution of Change from Baseline to Week 24 in ALSFRS-R Scores



5.3 Pharmacokinetic properties

Table 3: Summary of Edaravone Pharmacokinetic Parameters in ALS patients

Study #	Trial design	Dosage, route of administration and duration	Study subjects (n)	Mean age (Range)	Sex (%)
MCI186-19	Randomized, placebo-controlled, double-blind study	Intravenous infusion of 60 mg over 60 minutes, 6 months (6 cycles) treatment	Edaravone (69)	60.5 (30-75)	Male (55%) Female (45%)
			Placebo (68)	60.1 (38-75)	Male (60%) Female (40%)

	C_{max}	T_{max}	$t_{½}$ (h)	$AUC_{0-\infty}$	CL	Vd
Single dose mean	1046.6 ng/mL	1[h]	6.34 [h]	1362.3 ng*h/mL	43.7 [L/h]	80.9 [L]

Absorption: RADICAVA is administered by IV infusion. The maximum plasma concentration (C_{max}) of edaravone was reached by the end of infusion. There was a trend of a more than dose-proportional increase in area under the concentration-time curve (AUC) and C_{max} of edaravone. With multiple-dose administration, edaravone does not accumulate in plasma.

Distribution: Edaravone is bound to human serum proteins (92%), mainly to albumin, with no concentration dependence in the range of 0.1 to 50 micromol/L.

Metabolism: Edaravone is metabolized to a sulfate conjugate and a glucuronide conjugate, which are not pharmacologically active. The glucuronide conjugation of edaravone involves multiple uridine diphosphate glucuronosyltransferase (UGT) isoforms (UGT1A6, UGT1A9, UGT2B7, and UGT2B17) in the liver and kidney. In human plasma, edaravone is mainly detected as the sulfate conjugate, which is presumed to be formed by sulfotransferases.

Elimination: In Japanese and Caucasian healthy volunteer studies, edaravone was excreted mainly in the urine as its glucuronide conjugate form (70-90% of the dose). Approximately 5- 10% of the dose was recovered in the urine as the sulfate conjugate, and only 1% of the dose or less was recovered in the urine as the unchanged drug. *In vitro* studies suggest that the sulfate conjugate of edaravone is hydrolyzed back to edaravone, which is then converted to the glucuronide conjugate in the human kidney before excretion into the urine. The mean terminal elimination half-life of edaravone is 4.5 to 6 hours. The half-lives of its metabolites are 2 to 2.8 hours.

Special Populations and Conditions

Pediatrics: Safety and effectiveness of RADICAVA in pediatric patients have not been established.

Geriatrics: No age effect on edaravone pharmacokinetics has been found.

Sex: No gender effect on edaravone pharmacokinetics has been found.

Pregnancy and Breast-feeding: There are no adequate data on the developmental risk associated with the use of RADICAVA in pregnant women. There are no data on the presence of edaravone in human milk, the effects on the breastfed infant, or the effects of the drug on milk production.

Ethnic origin: There were no significant racial differences in C_{max} and AUC of edaravone between Japanese and Caucasian subjects.

Hepatic Insufficiency: Mild, moderate and severe hepatic impairment have no clinically significant effects on the pharmacokinetics of edaravone. No dose adjustment is needed in these patients.

Renal Insufficiency: Mild and moderate renal impairment have no clinically significant effects on the pharmacokinetics of edaravone. No dose adjustment is needed in these patients. No studies have been performed to characterize the pharmacokinetics of edaravone or establish its safety in patients with severe renal impairment.

5.4 Preclinical safety data

Carcinogenesis

The carcinogenic potential of edaravone has not been adequately assessed.

Mutagenesis

Edaravone was negative in *in vitro* (bacterial reverse mutation and Chinese hamster lung chromosomal aberration) and in vivo (mouse micronucleus) assays.

Impairment of Fertility

Intravenous administration of edaravone (0, 3, 20, or 200 mg/kg) prior to and throughout mating in males and females and continuing in females to gestation day 7 had no effect on fertility; however, disruption of the estrus cycle and mating behavior was observed at the highest dose tested. No effects on reproductive function were observed at the lower doses, which are up to 3 times the RHD of 60 mg, on a body surface area (mg/m^2) basis.

Reproduction

In rats, intravenous administration of edaravone (0, 3, 30, or 300 mg/kg/day) throughout the period of organogenesis resulted in reduced fetal weight at all doses. In dams allowed to deliver naturally, offspring weight was reduced at the highest dose tested. Maternal toxicity was also observed at the highest dose tested. There were no adverse effects on reproductive function in the offspring. A no-effect dose for embryofetal developmental toxicity was not identified; the low dose is less than the recommended human dose of 60 mg, on a body surface area (mg/m^2) basis.

In rabbits, intravenous administration of edaravone (0, 3, 20, or 100 mg/kg/day) throughout the period of organogenesis resulted in embryofetal death at the highest dose tested, which was associated with maternal toxicity. The higher no-effect dose for embryofetal developmental toxicity is approximately 6 times the RHD on a body surface area (mg/m^2) basis.

The effects on offspring of edaravone (0, 3, 20, or 200 mg/kg/day), administered by intravenous injection to rats from gestation day 17 throughout lactation, were assessed in two studies. In the first study, offspring mortality was observed at the high dose and increased activity was observed at the mid and high doses. In the second study, there was an increase in stillbirths, offspring mortality, and delayed physical development (vaginal opening) at the highest dose tested. Reproduction function in offspring was not affected in either study. Maternal toxicity was evident in both studies at all but the lowest dose tested. The no-effect dose for developmental toxicity (3 mg/kg/day) is less than the RHD on a mg/m^2 basis.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium bisulfite, L-cysteine hydrochloride hydrate, sodium chloride, sodium hydroxide, phosphoric acid

6.2 Incompatibilities

RADICAVA should be diluted with physiological saline (if the product is mixed with any infusion fluids including various saccharides, the concentration of edaravone may decrease with time).

RADICAVA should not be mixed with total parenteral nutrition preparations and/or amino-acid infusions before administration and should not be administered through the same intravenous line as those preparations (if the product is mixed with them, the concentration of edaravone may decrease with time).

RADICAVA should not be mixed with infusions of anticonvulsants including diazepam, phenytoin sodium, etc. (the solution may become cloudy).

RADICAVA should not be mixed with potassium canrenoate (the solution may become cloudy).

6.3 Shelf life

Please refer to expiry date on outer carton.

6.4 Special precautions for storage

Do not store above 30°C.

6.5 Nature and contents of container

Clear glass ampoule
Pack size: 10 x 20mL in a paper carton.

6.6 Special precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

Manufactured by:
Nipro Pharma Corporation, Ise Plant,
Mie, Japan

Packed and released by:
PT Mitsubishi Tanabe Pharma Indonesia
Bandung, Indonesia

Product Registration Holder:
Mitsubishi Tanabe Pharma Malaysia Sdn. Bhd.
Suite 8-3, Level 8, Wisma UOA Damansara II,
No. 6, Changkat Semantan, Damansara Heights,
50490 Kuala Lumpur, Malaysia

Date of Package Insert
01 February 2023 (version 3.0)